

PsyBuddy

Professional System Overview & Architecture

This document outlines the full architecture, modules, AI systems, and platform capabilities of PsyBuddy — a cross-platform AI-driven mental health intelligence system.

1. Product Vision

PsyBuddy is an **AI-driven mental health intelligence platform** that:

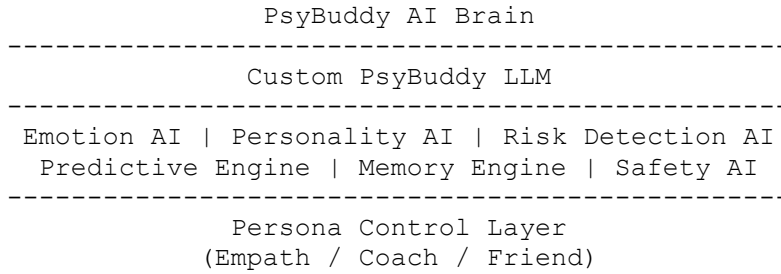
- Understands user psychology
- Predicts mental health risks
- Provides emotional and behavioural guidance
- Acts as a long-term cognitive companion

Core idea:

A **personal AI mental health ecosystem**, not just a Chabot.

2. Core AI Architecture

PsyBuddy AI Brain



3. Core System Components

1. Custom PsyBuddy LLM

Your proprietary model.

Purpose:

- Conversation intelligence
- Emotional reasoning
- Psychological insight
- Contextual memory usage

Recommended size:

1B – 3B parameters.

Capabilities:

- Therapy-style responses
- Context awareness
- Persona-controlled responses
- Long-term user modeling

4. Hybrid Cognitive AI Modules

Emotion Detection Engine

Detects:

- Stress
- Anxiety
- Sadness
- Mood fluctuations
- Emotional triggers

Input:

- Chat
- Journal
- Dream logs

Output:

```
emotion_score  
mental_state  
stress_index  
mood_timeline
```

Personality Detection Engine

AI analyzes long-term patterns.

Detects:

- Big Five personality traits
- Social behavior
- Motivation patterns
- Cognitive style
- Emotional stability

Updates continuously.

Psychological Risk Detection System

Detects early signals of:

- Depression
- Burnout

- Anxiety disorders
- Addiction tendencies
- Social withdrawal
- Behavioral instability

This is **very important for PsyBuddy's uniqueness.**

Predictive Mental Health Engine

This is one of the most advanced modules.

AI predicts:

- Burnout risk
- Depression onset
- Stress spikes
- Behavioral changes
- Mood crashes

How it works:

- Time-series behavioral analysis
- Pattern detection
- Emotional trajectory modeling

5. Persona Intelligence Layer

Triple Persona System

Empath

Role:

Emotional support.

Used when:

- Sadness
- Anxiety
- Crisis
- Emotional distress

Tone:

Therapeutic and calm.

Coach

Role:

Motivation and improvement.

Used when:

- Career stress
- Burnout risk
- Productivity problems
- Habit change

Tone:

Encouraging and structured.

Friend

Role:

Companion and casual interaction.

Used when:

- Normal mood
- Loneliness
- Daily conversation

Tone:

Relaxed and friendly.

6. Crisis Detection + Safety Layer

If the system detects risk:

Suicide signals
Severe depression
Extreme anxiety
Self-harm patterns

System response:

Persona forced → Empath
Safety Layer Activated
Risk Monitoring Increased

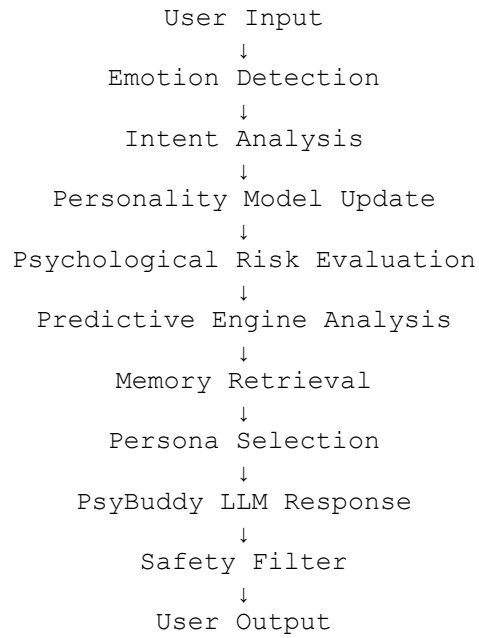
Safety actions:

- Gentle support responses
- Suggest professional help
- Crisis resources
- Monitoring escalation

This is critical for ethical AI.

7. AI Decision Pipeline

How PsyBuddy processes a message:



8. Memory Architecture

Emotional Memory

Stores:

- Emotional history
- Trigger events
- Mood trends

Semantic Memory

Stores:

- Important conversations
- User goals
- Life events

Behavioural Memory

Tracks:

- Activity changes
- Sleep signals
- Chat frequency
- Mood patterns

9. Psychological Testing System

Users can take tests such as:

- IQ tests
- Personality tests
- Aptitude tests
- Clinical screening tests

System features:

- Adaptive testing
- AI interpretation
- Integrated psychological profile
- Long-term tracking

Results are combined with AI predictions.

10. Cross-Platform Architecture

Supported platforms:

Mobile
Tablet
Desktop
Web

Technology stack:

Frontend:
Flutter

Backend:
FastAPI

AI Layer:
PyTorch + Transformers

Database:
MongoDB (cloud)
SQLite (offline)

Vector Memory:
FAISS

11. Offline Mode

When internet is unavailable:

Local AI modules work:

- Emotion detection
- Journaling analysis
- Basic conversation
- Memory storage

Local storage:

SQLite

When internet returns:

Sync engine activates.

12. Sync Engine

Synchronizes:

Chat history

Journals

Mood timeline

AI predictions

Psychological test results

Example:

Phone → Desktop → Web.

13. Notification System

Smart AI notifications.

Examples:

Mood reminders
Journal reminders
Mental health alerts
Stress warning
AI suggestions

Example:

"Your stress level has increased this week. Would you like to talk?"

14. Privacy & Security System

Critical for mental health apps.

Features:

End-to-end encryption

Secure storage

AI safety guardrails

Data anonymization

User data ownership

Optional local-only mode

15. PsyBuddy Dashboard (Final Version)

Main dashboard sections:

Home
AI Summary
Mental Health Status
Mood Trend
Recent Insights

AI Insight Panel

Shows:

Mental stability score
Burnout risk
Stress index
Personality insights

Prediction Panel

Shows:

Upcoming risk alerts
Mood forecast
Behavior trends

Journal & Dream Logs

Track:

- Emotional entries
- Dreams
- Reflection

Psychological Profile

Displays:

Personality analysis
Test results
AI evaluation

Persona Panel

User can switch between:

Empath
Coach
Friend
Auto Mode

16. Advanced Features (Final Product)

AI-powered personality evolution tracking

Mental health digital twin

Behavior prediction graphs

Emotion timeline visualization

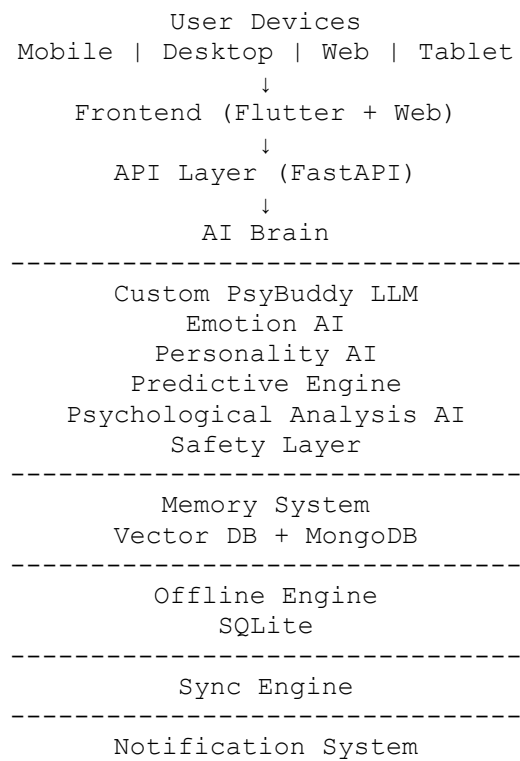
AI-generated therapy suggestions

Life pattern analysis

Mental stability score

Cognitive growth tracking

17. Final System Architecture



18. Development Roadmap

Phase 1 — Core Platform (2 months)

User system
Chat system
Journal system
Authentication
Database structure

Phase 2 — AI Foundation (2–3 months)

Emotion detection
Memory system
Persona system
Basic AI conversation

Phase 3 — Custom PsyBuddy LLM (3 months)

Dataset preparation
Model training
Fine-tuning
Deployment

Phase 4 — Advanced AI Modules (3 months)

Personality detection
Risk detection
Predictive engine

Phase 5 — Psychological Testing System (2 months)

Test engine
Result analysis
AI integration

Phase 6 — Safety + Clinical Layer (1 month)

Crisis detection
AI safety filters

Phase 7 — Cross Platform + Offline Sync (1–2 months)

Offline mode
Sync engine
Push notifications

Estimated Timeline

Full system:

12 – 14 months

But MVP:

6 months.

19. Final PsyBuddy Version (What It Becomes)

A platform that combines:

- AI therapist
- AI coach
- AI companion
- Predictive mental health system
- Psychological evaluation system

This is **far beyond typical mental health apps**.

Important (Strategic Insight)

If built properly, PsyBuddy could evolve into:

- Mental Health Operating System
- Clinical AI assistant
- Behavior prediction platform
- AI wellness ecosystem

Datasets to train on...

Empath

1. HuggingFace – Mental Health Counseling Conversations

https://huggingface.co/datasets/Amod/mental_health_counseling_conversations?

Real therapist-style Q&A conversations

Clean format: Context → Response

Fully downloadable via HuggingFace

Why it's good:

Real-world counseling tone

Already anonymized

Plug-and-play for fine-tuning

Format:

JSON / Parquet (easy for PyTorch / HF datasets)

2. MentalChat16K (Best Overall Dataset)

HuggingFace: <https://huggingface.co/datasets/ShenLab/MentalChat16K>

GitHub: <https://github.com/ChiaPatricia/MentalChat16K>

~16,000 conversations

Mix of:

Real anonymized therapy transcripts

Synthetic high-quality dialogues

Why it's powerful:

Covers depression, anxiety, grief

Designed specifically for training empathetic LLMs

Open license (CC BY-SA)

Format:

JSON conversational format

3. Kaggle – Mental Health Counseling Dataset

<https://www.kaggle.com/datasets/melissamonfared/mental-health-counseling-conversations-k?>

Therapy Q&A scraped from public counseling platforms

Lightweight dataset

Why use it:

Very easy to start with

No preprocessing headache

Format:

CSV (great for quick experiments)

4. KokoroChat (High-Quality Simulated Therapy Dialogues)

GitHub: <https://github.com/UEC-InabaLab/KokoroChat>

What it is:

~6,500 full conversations

Created by trained counselors via role-play

Why it's valuable:

Very realistic structure

Safer than real patient data

Multi-turn conversations (rare!)

5. Cactus (CBT-Based Conversations)

 Paper + Code (open): <https://arxiv.org/abs/2407.03103>

What it is:

Structured conversations using CBT therapy techniques

Why it's unique:

Teaches how therapy works, not just tone

6. CounselChat Dataset (Real Therapist Responses)

GitHub: <https://github.com/nbertagnolli/counsel-chat>

What you get:

Real Q&A between users and licensed therapists

Scraped from a public counseling platform

Why useful:

High-quality expert-written responses

Good for training:

empathy

safe responses

✓ Easy to load via HuggingFace (load_dataset)

7. AnnoMI (Motivational Interviewing Dataset)

GitHub: <https://github.com/uccollab/AnnoMI>

What you get:

133 full therapy conversations

Expert annotated

Why it's powerful:

Teaches therapy techniques (Motivational Interviewing)

Includes:

good vs bad therapist behavior

Fully open-source

High-quality annotations

8. Psy-Insight (Advanced Multi-Turn Counseling Dataset)

GitHub: <https://github.com/ckqqqq/Psy-Insight>

What you get:

~6,000 multi-turn dialogues

Rich annotations:

emotions

therapy strategies

reasoning steps

Why it's powerful:

Teaches:

how therapists think

not just what they say

Great for advanced LLM fine-tuning

9. MESC (Emotional Support Conversations)

GitHub: <https://github.com/chuyq/MESC>

<https://www.kaggle.com/datasets/rahulbaburaj/annomi>

What you get:

Therapy-style conversations from In Treatment

Includes:

text

audio/video cues

Why useful:

Very realistic therapy sessions

Emotion-rich dialogues

Downloadable dataset

10. CaiTI Dataset (CBT + Therapy Conversations)

GitHub: https://github.com/Columbia-ICSL/CaiTI_dataset

What you get:

Motivational Interviewing + CBT conversations

Curated by therapists

Good for:

structured therapy reasoning

Coach

1. Health Coaching Conversations (Closest to Real Coaching)

GitHub: <https://github.com/uic-nlp-lab/virtualcoachdata>

What it is:

Real conversations between health coaches and patients via SMS

Includes:

Goal setting

Follow-ups

Behavior change coaching

Why this is GOLD:

True 1-on-1 coaching dynamics

Uses SMART goals (exactly what mentors do)

Multi-turn, realistic conversations

Key insight:

Weekly coaching loops (goal → progress → feedback)

Perfect for building your Coach persona in PsyBuddy

This is the closest thing to:

“mentor guiding a person step-by-step”

2. Google Education Dialogue Dataset (Teacher = Mentor)

GitHub: <https://github.com/google-research-datasets/Education-Dialogue-Dataset>

What it is:

~40,000 full conversations

Teacher ↔ Student interactions

Includes:

Learning preferences

Adaptation strategies

Feedback loops

Why it's powerful:

Mentorship = teaching + adaptation

Includes personalization signals

Structure:

Multi-turn dialogue (10–15 turns per convo)

Metadata about learning style

✓ Very strong for:

AI tutors

Skill coaching agents

Adaptive mentorship

3. CCPE (Coached Preference Elicitation)

GitHub: <https://github.com/google-research-datasets/ccpe>

What it is:

~500 dialogues (~12k utterances)

“Assistant” helps user refine preferences

Why it matters:

Teaches:

Asking the right questions

Guiding decisions

Clarifying goals

This is mentorship behavior, not just chat

Friend

1. Topical-Chat (Human ↔ Human Conversations)

GitHub: <https://github.com/alexa/Topical-Chat>

What it is:

Real conversations between two people

Topics:

movies

sports

hobbies

general interests

Why useful:

Feels like:

friends discussing topics

casual but informative

Multi-turn + natural flow

2. Cornell Movie Dialogs Corpus (Classic)

Download: https://www.cs.cornell.edu/~cristian/Cornell_Movie-Dialogs_Corpus.html

What it is:

220k+ lines of dialogue from movies

Why it works:

Many conversations = friend-like banter

Includes:

sarcasm

humor

emotional exchanges

Widely used for chatbot training

3. OpenSubtitles Dataset (Massive Scale)

Available via: <https://opus.nlpl.eu/OpenSubtitles>

What it is:

Millions of conversations from subtitles

Why powerful:

Extremely diverse:

casual talk

arguments

friendships

relationships

Best for:

Large-scale pretraining

4. Persona-Chat (Friend + Personality Simulation)

<https://parl.ai/projects/personachat/>

What it is:

Conversations where each speaker has a persona

Why useful:

Simulates:

friends with personalities

consistent behavior

Perfect for:

building relatable AI personalities

5. Blended Skill Talk (Meta AI)

https://parl.ai/projects/blended_skill_talk/

What it is:

~7,000 conversations

Combines:

empathy

personality

knowledge

Designed for:

engaging, human-like conversation

6. LMSYS Chat 1M (Modern Real Conversations)

<https://huggingface.co/datasets/lmsys/lmsys-chat-1m>

What it is:

1 million real-world conversations

Why useful:

Very natural language

Real user style (casual, messy, human)

Great for:

modern conversational tone

7. Everyday Conversations (Simple Friendly Chats)

<https://huggingface.co/datasets/HuggingFaceTB/everyday-conversations-llama3.1-2k>

What it is:

Short, simple dialogues

Greetings + small talk

Good for:

lightweight models

Emotion - Speech

1. RAVDESS (★ MUST START HERE)

Download: <https://zenodo.org/records/1188976>

What it is:

7,000+ audio/video files

24 actors speaking with emotions

Emotions:

happy, sad, angry, fear, disgust, surprise, calm, neutral

Why it's gold:

Clean, labeled, high-quality audio

Multiple intensity levels

Details:

48kHz WAV audio

Fully open (Creative Commons)

Best for:

Speech Emotion Recognition (SER) baseline

Feature extraction (MFCC, spectrograms)

2. CREMA-D (Crowd-Sourced Emotional Speech)

Download: <https://github.com/CheyneyComputerScience/CREMA-D>

What it is:

7,000+ audio clips

Real human emotional expressions

Why useful:

More natural than RAVDESS

Includes:

different accents

crowd-sourced labelling

Good for:

Generalization (real-world voice AI)

3. SAVEE (Surrey Audio-Visual Expressed Emotion)

Download: <https://kahlan.eps.surrey.ac.uk/savee/>

What it is:

Male speakers

British English

Good for:

accent diversity

4. EmoGator (Very Unique)

GitHub: <https://github.com/fredbuhl/EmoGator>

What it is:

32,000+ audio samples

Vocal bursts:

laughter

sighs

crying

Why important:

Captures non-verbal emotion signals

This is something most models miss

5. MELD (Multi-Party Emotion Conversations)

Download: <https://affective-meld.github.io/>

What it is:

Conversations from TV show Friends

Multi-speaker dialogue

Includes:

audio

text

emotion labels

Best for:

conversational emotion detection

6. SER Dataset Collection (77+ datasets)

GitHub: <https://github.com/SuperKogito/SER-datasets>

What it is:

Curated list of 70+ datasets

Direct download links

Best if you want:

scale + variety

7. Combined SER Dataset Repo

GitHub: https://github.com/standing-o/Combined_Dataset_for_Speech_Emotion_Recognition

Merges:

RAVDESS

CREMA-D

others

Emotion – Video

1. RAVDESS (Audio + Video Emotion Dataset) ★ MUST USE

Download: <https://zenodo.org/records/1188976>

What it is:

7,356 audio + video files

24 actors expressing emotions

Includes:

Video (facial expressions)

Audio (speech tone)

Emotions:

happy, sad, angry, fear, disgust, surprise, calm

Why it's important:

One of the most widely used multimodal emotion datasets

Clean + labeled + easy to train

Fully downloadable

Great for baseline multimodal models

Fact: Contains thousands of audio-visual recordings under controlled conditions

2. MELD (★ BEST for Conversation + Emotion)

Download: <https://affective-meld.github.io/>

What it is:

Conversations from the TV show Friends

Multi-speaker dialogues

Includes:

Video

Audio

Text transcripts

Emotion labels

Scale:

~13,000 utterances across dialogues

Why it's gold:

Real multi-turn conversations + emotions

Context-aware emotion detection

Best dataset for:

Conversation retrieval + emotion detection together

3. IEMOCAP (Semi-open but VERY important)

<https://sail.usc.edu/iemocap/>

What it is:

~12 hours of video + audio conversations

Includes:

Video

Audio

Transcripts

Emotion labels (categorical + valence/arousal)

Features:

Dyadic (2-person conversations)

Motion capture + facial tracking

Extremely valuable for:

conversational emotion modeling

Fact: Contains multimodal dyadic interactions with detailed annotations

Requires request (but widely used in research)

4. DFEW (Dynamic Facial Expression in-the-Wild)

<https://dfew-dataset.github.io/>

What it is:

16,000+ video clips from movies

Includes:

Real-world video scenes

Emotion annotations

Features:

7 emotion categories

Multiple annotators

Best for:

real-world robustness (not staged data)

Fact: Contains 16,372 challenging video clips with labeled emotions

5. CMU-MOSEI (★ MULTIMODAL + TEXT + VIDEO)

<https://github.com/A2Zadeh/CMU-MultimodalSDK>

What it is:

One of the largest multimodal datasets

Includes:

Video (facial expressions)

Audio

Text transcripts

Sentiment + emotion

Why powerful:

Real YouTube videos

Natural conversations

Best for:

emotion + sentiment + context understanding

6. EEV (Evoked Expressions from Videos)

<https://github.com/google-research-datasets/eev>

What it is:

23,000+ videos

36M+ emotion annotations

Includes:

Viewer emotional reactions to videos

Scale:

~1,700 hours of content

Best for:

large-scale emotion modeling

7. VEATIC (Context-Aware Emotion in Video)

<https://veatic.github.io>

What it is:

Video clips with continuous emotion tracking

Includes:

Frame-level:

valence

arousal

Best for:

temporal emotion understanding

Fact: Includes continuous emotion annotations across video frames

8. HEU Emotion Dataset (Large-Scale Multimodal)

(Available via research repo)

What it is:

~19,000 video clips

Real-world + curated sources

Includes:

facial expressions

body posture

speech

Best for:

full multimodal emotion AI

Fact: Contains thousands of clips with multiple modalities and 10 emotions

Emotion – Text

1. GoEmotions (★ BEST OVERALL)

<https://github.com/google-research/google-research/tree/master/goemotions>

What it is:

58k Reddit comments

27 emotion labels + neutral

Why it's powerful:

Fine-grained emotions (not just happy/sad)

One of the largest labeled datasets

Fact: Designed as a large fine-grained emotion dataset

Best for:

Advanced emotion detection models

2. HuggingFace “Emotion” Dataset (dair-ai)

<https://huggingface.co/datasets/dair-ai/emotion>

What it is:

Twitter text labeled with 6 emotions:

anger, fear, joy, love, sadness, surprise

Clean + preprocessed dataset

Best for:

Quick training

Baseline models

3. Kaggle Emotion Dataset (Twitter-based)

<https://www.kaggle.com/datasets/parulpandey/emotion-dataset>

What it is:

Social media text

Same 6 emotion categories

Good for:

scaling data

combining with HF dataset

Based on labeled tweets with emotional categories

4. Sentiment + Emotion Multi-task Dataset

<https://www.kaggle.com/datasets/kushagra3204/sentiment-and-emotion-analysis-dataset>

What it is:

Emotion + sentiment labels

Best for:

multi-task learning

building smarter models

5. Text Emotion Classification Dataset (Generic)

<https://www.opendatabay.com/data/ai-ml/afdefbf4-147a-44ad-9a6b-1e8e584c3542>

What it is:

Text snippets labeled with emotions

Useful for:

experimentation

dataset augmentation

Designed for NLP emotion classification tasks

6. DailyDialog (★ MUST USE)

<http://yanran.li/dailydialog.html>

What it is:

Multi-turn conversations

Includes:

emotion labels

dialogue acts

Best for:

conversation retrieval

dialogue modeling

Widely used for multi-turn emotion understanding

7. EmotionLines (Conversation Emotion Dataset)

<https://github.com/declare-lab/EmotionLines>

What it is:

Multi-party conversations

Emotion labeled per utterance

Includes:

~29,000 sentences across conversations

Best for:

contextual emotion detection

8. EmoryNLP Dataset (Friends TV Conversations)

<https://github.com/emorynlp/emotion-detection>

What it is:

Conversations from Friends

Emotion labels per dialogue

~12,000 utterances across episodes

Best for:

realistic conversational emotion

9. EmoWOZ (Task-Oriented Conversations + Emotion)

<https://github.com/emowoz/EmoWOZ>

What it is:

11k+ dialogues

Emotion labels in task-based conversations

83k+ emotion annotations

Best for:

assistant-style conversations

10. EmpatheticDialogues (★ VERY IMPORTANT)

<https://github.com/facebookresearch/EmpatheticDialogues>

What it is:

25k conversations grounded in emotional situations

Best for:

empathy + response generation

11. The Pile (Massive Pretraining Dataset)

<https://pile.eleuther.ai>

What it is:

800GB+ open text corpus

One of the largest open datasets for LLM training

Best for:

pretraining

embedding models

12. Reddit / Open Conversation Datasets

<https://github.com/PolyAI-LDN/conversational-datasets>

What it is:

Reddit threads

forum conversations

Best for:

real-world conversational tone